

Principles of Computer Security, Sixth Edition

Chapter 13 Solutions

Key Terms Quiz

1. protocol analyzer
2. false positive
3. honeypot or honeynet
4. content-based signature
5. anomaly detection model
6. Port mirroring
7. analysis engine
8. Banner grabbing
9. Context-based signature
10. Suricata

Multiple-Choice Quiz

1. **A.** The two main types of intrusion detection systems are *network based* and *host based*. Network-based systems monitor network connections for suspicious traffic. Host-based systems reside on an individual system and monitor that system for suspicious or malicious activity.
2. **B.** The two main types of IDS signatures are *context based* and *content based*. Context-based signatures examine traffic and how that traffic fits into the other traffic around it. A port scan is a good example of a context-based signature. A content-based signature looks at what is inside the traffic, such as the contents of a specific packet.
3. **D.** A passive, host-based IDS runs on the local system, cannot interfere with traffic or activity on that system, and would have access to local system logs.
4. **B.** A network-based IDS typically cannot decrypt and read encrypted traffic. This is one of the principal weaknesses of network-based intrusion detection systems.

5. **C.** An active IDS can perform all the functions of a passive IDS (monitoring, alerting, reporting, and so on), with the added ability of responding to suspected attacks with capabilities such as sending TCP reset messages to the source and destination IP addresses.
6. **A.** Honeypots are designed to attract attackers by providing what appear to be easy, inviting targets. The honeypot collects and records the activity of attackers and their tools.
7. **C.** Banner grabbing is the sending of a protocol across a known port and examining the return signal as to the protocol in order to determine the version of the service that is responding.
8. **B.** Preventative IDSs are designed to stop malicious activity from occurring.
9. **C.** IPS stands for intrusion prevention system.
10. **D.** A protocol analyzer is a very flexible tool and can be used for network traffic analysis, statistics collection, and monitoring and identification of suspicious or malicious traffic.

Essay Quiz

1. The response for the initial portion of this question, “discuss the differences between an anomaly-based and a misuse-based detection model,” should include the following elements:

Anomaly-based models:

- Tend to be more complex than misuse based
- Must know what “normal” behavior is in order to classify abnormal behavior
- Must have the ability to adjust the “normal behavior” baseline to accommodate new systems and applications
- Tend to have a high number of false positives
- Typically are not restricted to just a specific set of signatures

Misuse-based models:

- Are easier to implement and more common
- Look for known patterns of suspicious or malicious activity
- Are very reliant on a developed signature set

- Will likely not see a new attack if there is no signature to match it

The remaining two questions are essentially opinion based, and the answer should explain the choices between the two models and provide supporting details as to why they would choose one model over the other.

2. The key items in the chapter are intrusion detection systems, both host and network based. These can be compared and contrasted. There are also intrusion prevention systems, and these can be compared and contrasted to the IDS types already mentioned. The third major element is honeypots/nets.